

PRODUCTION MICROSERVICES BLUEPRINT

Full-Stack Enterprise Architecture Specification & Cloud-Native Topology

PRESENTATION LAYER (FRONTEND)

Web Application

React 19 + TypeScript

- Next.js 15 (SSR / SSG Architecture)
- Tailwind CSS 4 (Utility-first Styling)
- TanStack Query v5 (Server State Management)
- Zustand (Global Client State Management)
- Axios Client (HTTP REST Operations)
- React Hook Form + Zod (Schema Validation)

Mobile Application

Flutter 3.x / React Native

- Dio HTTP Client (Advanced Networking)
- Riverpod (Reactive State Management)
- Hive (High-performance Local Storage)
- Firebase Cloud Messaging (Push Notifications)

Frontend Testing Suite

- Vitest & React Testing Library (Unit & Component)
- Cypress E2E (End-to-End Automation Testing)

Frontend Security Policy

- **JWT Storage:** Strictly persisted within `HttpOnly` Cookies to mitigate XSS risks.
- **Axios Interceptors:** Implements automated request modification for token refreshing workflows.
- **CSRF Protection:** Synchronized multi-token handling patterns across asynchronous payloads.

Third-Party Integrations

- **Payments:** Stripe APIs & secure SDK integration hooks.
- **Comms:** SendGrid / Mailgun transactional routing.
- **Identity Providers:** Google & Facebook OAuth2 integrations.
- **Assets:** AWS S3 / Cloudinary cloud asset delivery networks.

Edge Network Constraints: HTTPS / TLS 1.3 + HSTS Explicit Policies | Content-Type: application/json | Authorization: Bearer <JWT_TOKEN> | CORS: Strict White-listed Origin Checks

API GATEWAY LAYER (SPRING CLOUD GATEWAY)

Spring Cloud Gateway 4.x Kernel

Dynamic Routing Matrix

- `/api/auth/**` → auth-service:8081
- `/api/users/**` → user-service:8082
- `/api/products/**` → product-service:8083

Security Guardrails

- Global Rate Limiting (Bucket4j + Redis)
- Real-time IP Black/Whitelisting
- Dynamic API Key Validation & Size Limits

Performance Optimization

- Reactive Response Caching & Collapsing
- Core HTTP Connection Pooling & GZip
- Dedicated WebSocket Stream Routing

System Resilience

- Resilience4j Circuit Breaker implementation
- Exponential Backoff Retries & Safe Fallbacks
- Strictly enforced Timeout thresholds (30s)

Architecture Observability

- End-to-End Tracing (Micrometer + Zipkin)
- Structured Logging & Prometheus Core Metrics
- Reactive Cluster Health Endpoint Checkers

API Control & Lifecycle

- Explicit Versioning (`/api/v1`, `/api/v2`)
- Unified Aggregated Swagger Documentation
- Automated Specification Contracts

CORE MICROSERVICES CLUSTER ARCHITECTURE

AUTH SERVICE [Port 8081]

1. INTERCEPTOR FILTER CHAIN

- ① Bucket4j + Redis (100 r/m/IP)
- ② JWT 0.12.x RSA 256 Verify
- ③ Spring Security Context Auth

2. CONTROLLER INTERFACE

- POST /login
- POST /register
- POST /refresh
- POST /logout
- GET /oauth2/*

3. CORE BUSINESS SERVICES

- AuthService (BCrypt, SendGrid)
- TokenService (JWT Generation / Validation via Redis storage engines)

4. PERSISTENCE & MAPPING

- Spring Data JPA + Hibernate 7.x
- MapStruct 1.6 DTO Layering

USER SERVICE [Port 8082]

1. INTERCEPTOR FILTER CHAIN

- ① Rate Limiting (100 r/m/IP)
- ② Token Security Parsing
- ③ ACL Context Binding

2. CONTROLLER INTERFACE

- GET /users
- POST /users
- PUT /users/{id}
- DELETE /users/{id}
- GET /users/me

3. CORE BUSINESS SERVICES

- UserService (@Transactional execution abstraction)
- @Cacheable / @CacheEvict routines

4. PERSISTENCE & MAPPING

- JPA + Custom Native SQL queries
- MapStruct DTO translation maps

PRODUCT SERVICE [Port 8083]

1. INTERCEPTOR FILTER CHAIN

- ① High-throughput (200 r/m/IP)
- ② JWT Authentication Layer
- ③ PreAuthorize Policy Mapping

2. CONTROLLER INTERFACE

- GET /products
- POST /products
- PUT /products
- DELETE /products
- GET /search

3. CORE BUSINESS SERVICES

- Product & Category Management
- Cluster Cache Evictions
- Elasticsearch index synchronizations

4. PERSISTENCE & MAPPING

- Criteria API & Specification Filters
- QueryDSL 5.x dynamic querying

PERSISTENCE, SHARED CACHING & ASYNCHRONOUS EVENT BROKER BUS

PostgreSQL 17 (Primary)

RELATIONAL TABLES MATRIX

- users, roles, permissions
- products, categories, reviews
- orders, audit_logs, refresh_tokens

PERFORMANCE OPTIMIZATION

- B-Tree Indexes (PK, FK constraints)
- GIN (Optimized Full-Text Search)
- BRIN (Time-series log analysis)

TOOLING SUITE

- Flyway 10.x Database Migrations
- pgBouncer Layered Connection Pools

Redis 7.x (Cache & Cache Store)

ACTIVE KEY STRATEGIES

- users::123 (User JSON, TTL: 30m)
- tokens::abc (JWT Store, TTL: 5m)
- products::456 (Data cache, TTL: 2h)
- rate_limit::IP (Counters, TTL: 1m)
- sessions::xyz (Session data, TTL: 15m)

DRIVER & TOOLING INFRASTRUCTURE

- Lettuce 6.x Reactive Cluster Driver
- Spring Data Redis Abstractions

RabbitMQ 4.x (Message Broker)

TOPOLOGY & EXCHANGES

- **user.exchange** (created, updated, deleted)
- **product.exchange** (created, updated)
- **order.exchange** (order.placed event topologies)

TARGET QUEUES

- email.queue
- notification.q
- audit.queue
- dlq.queue (Dead Letter)

ASYNCHRONOUS CONSUMERS

- Email Service & SendGrid Router
- Push Notifications Engine (FCM)
- Security Audit Logging Daemon

UNIFIED ARCHITECTURE OBSERVABILITY & TELEMETRY PLATFORM

Metrics Architecture

- Micrometer 1.14 Core Binder
- Prometheus Time-Series Scraper
- Grafana Dashboard Visualizations
- JVM Diagnostics: Heap, Thread states, GC activity, CPU usage bounds.
- Active Alert Routing: Slack Webhooks, Email Alert, PagerDuty pipelines.

Aggregated Enterprise Logging

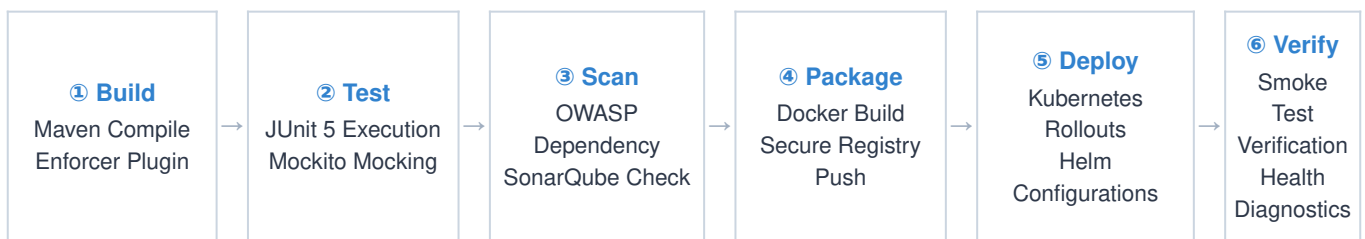
- SLF4J 2.x API & Logback 1.5 Framework
 - Logstash Encoder formatting JSON packets
 - Enterprise ELK Stack & CloudWatch storage
- Log Pattern Specifications:**
[TIMESTAMP] [LEVEL] [TRACE_ID]
[SERVICE] [CLASS] - MESSAGE

Trace & Audit Infrastructures

- Distributed Micrometer Tracing Core
- Brave / Zipkin Context Propagators
- OpenTelemetry Protocol Agnostic Standards
- Auditing: Spring Application Event models tracking system mutations, failed auth attempts, rate limits.

CONTINUOUS INTEGRATION & CONTAINER ORCHESTRATION TOPOLOGY

Pipeline Orchestration Automation (GitHub Actions / Jenkins)



Local Orchestration Container Isolation

Docker Compose Blueprint Engine

- API Gateway Network Host Configured at port :8080
- Core Autonomous Business Microservices isolated sequentially across :8081 , :8082 , and :8083 .
- Dedicated underlying backing stores instantiated uniformly on local daemon bindings: PostgreSQL :5432 and Redis :6379 .

Kubernetes Production Cluster Schematics

Namespace Config Strategy: **production**

- **Ingress Nginx Router**: Upstream TLS management and edge certificate bindings.
- **ClusterIP Services**: Provides rigid internal cluster pod mapping and service-discovery load balancer layers.
- **Deployments Strategy**: Enforces 3 separate active atomic Pod Replicas utilizing non-disruptive Zero-Downtime Rolling Updates.
- **Secrets/ConfigMaps Matrix**: Internal configuration parsing alongside runtime dynamic encryption providers via external Vault integrations.